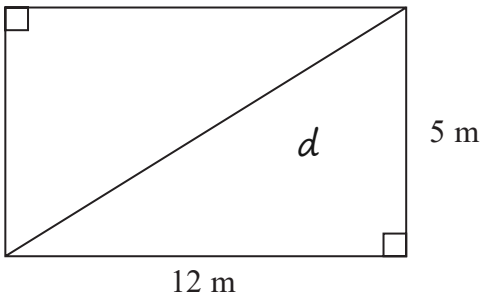
5 This rectangular frame is made from 5 straight pieces of metal.



The weight of the metal is 1.5 kg per metre.

Work out the total weight of the metal in the frame.

Diagonal d is the hypotenuse.

$d^2 = 12^2 + 5^2 = 144 + 25 = 169$

1 mark
Start Pythagoras

$d = \sqrt{169} = 13$ m

1 mark
Complete Pythagoras

Total length = 12 + 12 + 5 + 5 + 13 = 47 m

1 mark
Include all 5 pieces

Use 1.5 kg for every metre.

m	kg
1	1.5
47	70.5

(×47 from 1 to 47; ×47 from 1.5 to 70.5)

1 mark
Multiply total length by 1.5 kg per metre

Weight = 47 × 1.5 = 70.5 kg

1 mark
Correct weight

70.5

..... kg

(Total for Question 5 is 5 marks)